

ASSESSMENT TOOL 3- DATA VISUALIZATION DESIGN ACTIVITY

Course Code: DSA0504	Course Title: Query Processing for Data Science
CO Assessed: CO5	Assessment: Tool 3– Data Visualization Design Activity
Weightage: 25	Total Marks: 25 Marks
CO4 Statement: Create meaningful visualizations using Pandas and Matplotlib, including charts,distributions, relationships, and interactive visual representations for effective communication.	
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Section / Batch:2024	Date of Submission:24/8/26
Faculty In-charge: Dr. R.Mahesh Muthulakshmi	Project Mode: Individual Submission

QUESTION 1 – Autocorrelation vs Stable Bootstrap Estimates

A researcher concludes that a dataset is suitable for forecasting because it exhibits strong autocorrelation. Another researcher argues that stable bootstrap estimates are more important than autocorrelation for reliable prediction. Compare both positions and defend the stronger argument.

Objective

To determine whether strong autocorrelation alone is sufficient for reliable forecasting and to compare it with the importance of stable bootstrap estimates. An electricity-demand dataset and a Random Forest model are used.

Problem Statement

Strong autocorrelation indicates that observations are related across time, but it does not automatically guarantee accurate future predictions. The analysis compares temporal dependence with model stability and validation.

Dataset

Feature	Description
Previous_Demand	Electricity demand in the previous period
Demand_Lag2	Electricity demand two periods earlier
Temperature	Temperature during the period
Day_Type	0 = one day type, 1 = another day type
High_Demand	Target: 1 = High, 0 = Low

Machine Learning Model

Random Forest Classifier with 100 trees is used to predict whether electricity demand is High or Low. The data is divided into 75% training and 25% testing data.

Python Code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (
    accuracy_score, precision_score,
    recall_score, f1_score, confusion_matrix
)

np.random.seed(42)
n = 250
demand = []
value = 500

for i in range(n):
    value = 0.75 * value + np.random.normal(0, 40)
    demand.append(value)

df = pd.DataFrame({
    "Previous_Demand": demand,
    "Temperature": np.random.uniform(20, 40, n),
    "Day_Type": np.random.randint(0, 2, n)
})

df["Demand_Lag2"] = df["Previous_Demand"].shift(2)
df["High_Demand"] = (
    df["Previous_Demand"] > df["Previous_Demand"].median()
).astype(int)
df.dropna(inplace=True)

X = df[["Previous_Demand", "Demand_Lag2",
        "Temperature", "Day_Type"]]
y = df["High_Demand"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42, stratify=y
)

model = RandomForestClassifier(
    n_estimators=100, random_state=42
)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Accuracy :", accuracy_score(y_test, y_pred))
print("Precision:", precision_score(y_test, y_pred))
print("Recall   :", recall_score(y_test, y_pred))
print("F1 Score :", f1_score(y_test, y_pred))

cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:")
print(cm)
```

Visualizations

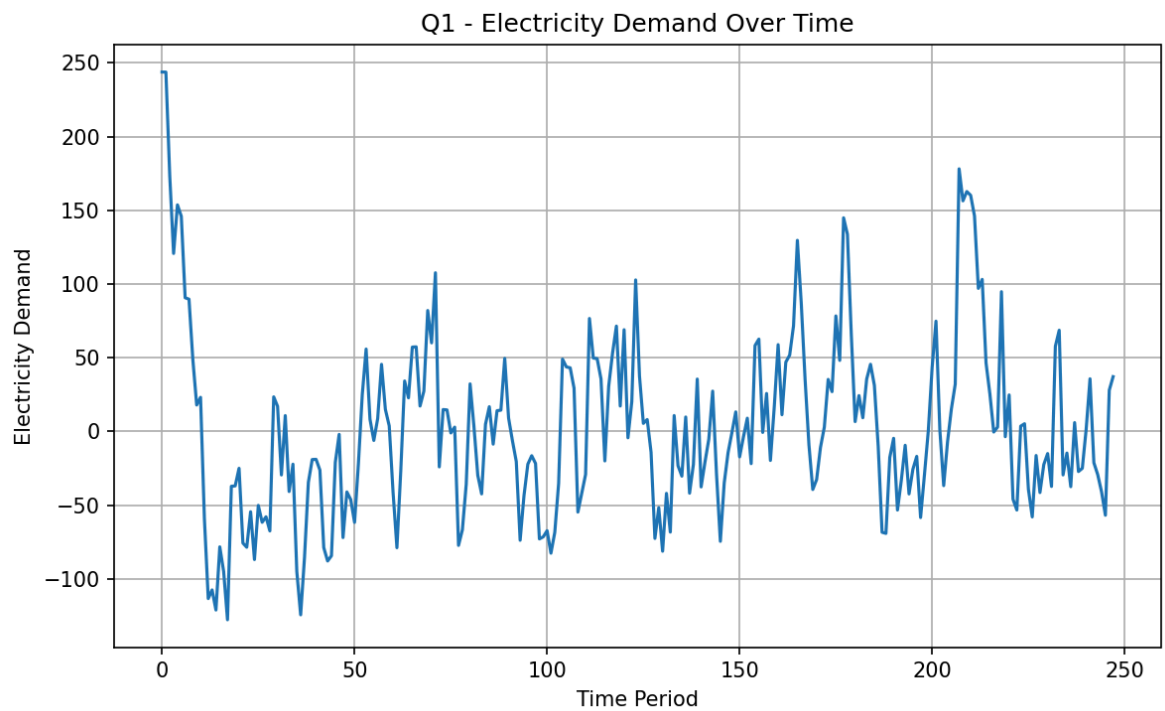


Figure 1. Electricity Demand Over Time

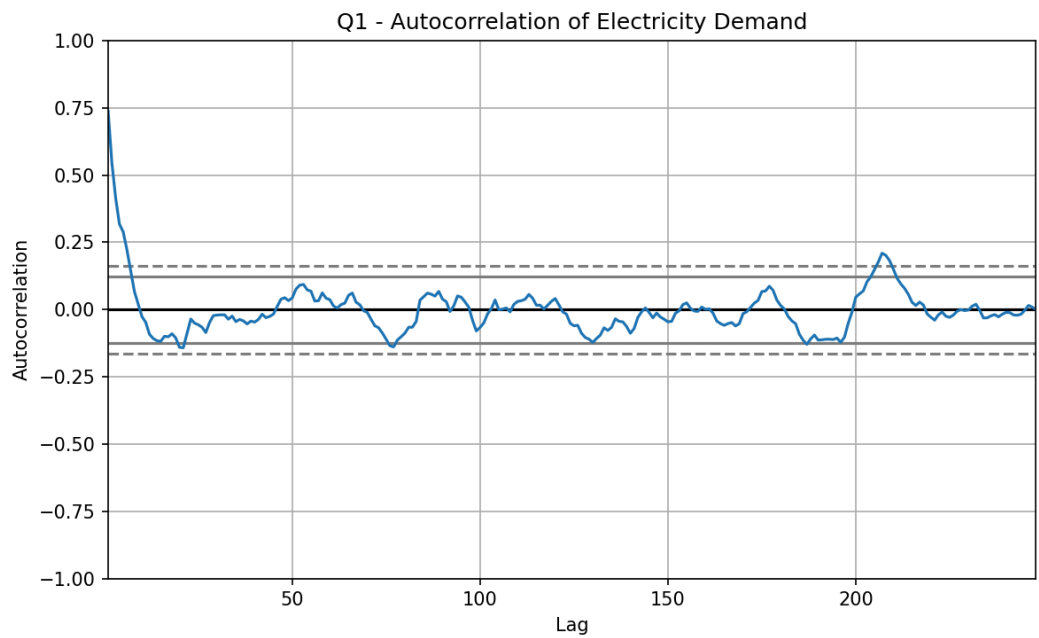


Figure 2. Autocorrelation of Electricity Demand

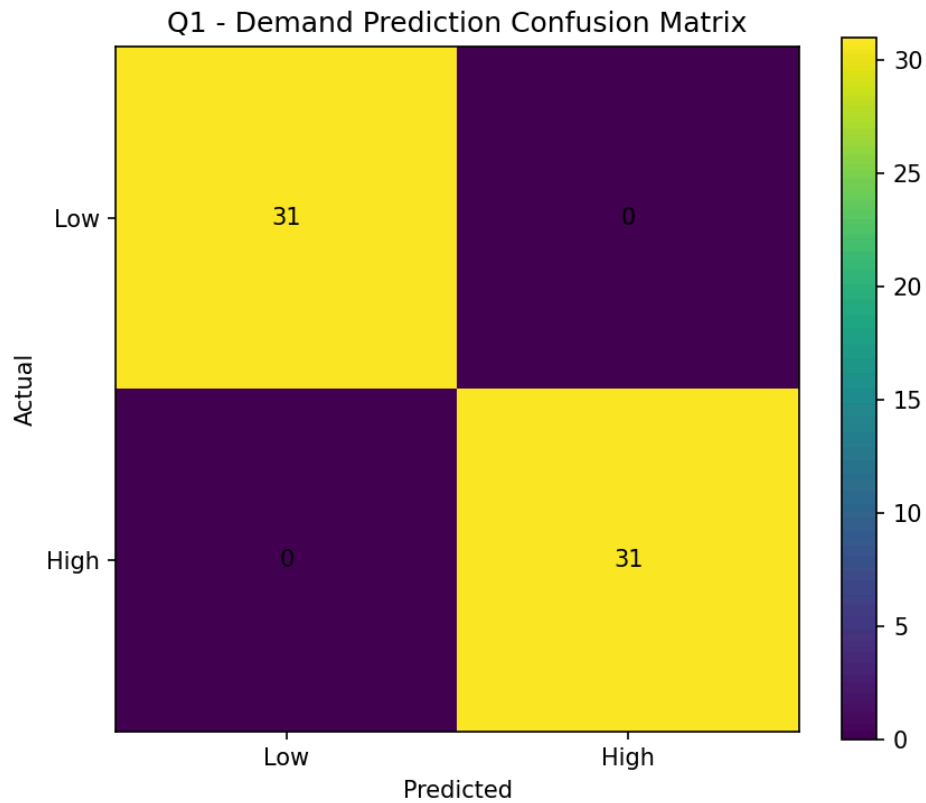


Figure 3. Demand Prediction Confusion Matrix

Evaluation Values

Parameter	Value
TP	31
TN	31
FP	0
FN	0
Accuracy	100.00%
Precision	100.00%
Recall	100.00%
F1 Score	100.00%

Calculation of Parameters

Accuracy = (TP + TN) / (TP + TN + FP + FN)

= (31 + 31) / (31 + 31 + 0 + 0) = 62 / 62 = 1.00

Accuracy = 100.00%

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) = 31 / (31 + 0) = 1.00 = 100.00\%$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) = 31 / (31 + 0) = 1.00 = 100.00\%$$

$$\text{F1 Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

$$= 2 \times (1.00 \times 1.00) / (1.00 + 1.00) = 1.00 = 100.00\%$$

Critical Evaluation

- Strong autocorrelation indicates temporal dependence and can support the use of forecasting methods.
- Autocorrelation alone does not prove that future predictions will be reliable; trends, seasonality and non-stationarity can create misleading patterns.
- Bootstrap or repeated resampling can show whether model performance remains stable across different samples.
- Stable validation results provide stronger evidence of reliability because they address uncertainty and sensitivity to the sample.
- Therefore, the second researcher's position is stronger, although autocorrelation remains useful for identifying temporal structure.

Conclusion

Autocorrelation is evidence of temporal dependence, not proof of predictive reliability. Reliable forecasting should combine temporal analysis with appropriate validation and stability assessment.

QUESTION 2 – Investment Decision Based on Predictive Analytics

A company must decide whether to invest millions of rupees based on predictive analytics derived from this dataset. Construct a decision framework that evaluates the reliability, limitations, and business risks associated with each visualization before making the investment decision.

Objective

To evaluate the reliability, limitations and business risks of predictive analytics before a high-value investment decision.

Problem Statement

A company plans to invest millions of rupees based on a loan-approval prediction system. Incorrect approvals or rejections can create financial losses, so the model and its visualizations must be evaluated before investment.

Dataset

Feature	Description
Applicant_Income	Applicant income
Loan_Amount	Requested loan amount
Credit_Score	Applicant credit score
Employment_Years	Years of employment
Debt_to_Income	Debt-to-income ratio
Loan_Approved	1 = Approved, 0 = Rejected

Machine Learning Model

Random Forest Classifier with 100 trees is used to predict loan approval.

Python Code

```
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (
    accuracy_score, precision_score,
    recall_score, f1_score, confusion_matrix
)

df = pd.read_csv("loan_approval_dataset.csv")

X = df.drop("Loan_Approved", axis=1)
y = df["Loan_Approved"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25,
    random_state=42, stratify=y
)

model = RandomForestClassifier(
    n_estimators=100, random_state=42
)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)

print("Accuracy :", round(accuracy, 4))
print("Precision:", round(precision, 4))
print("Recall   :", round(recall, 4))
print("F1 Score :", round(f1, 4))

cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:")
print(cm)
```

Visualizations

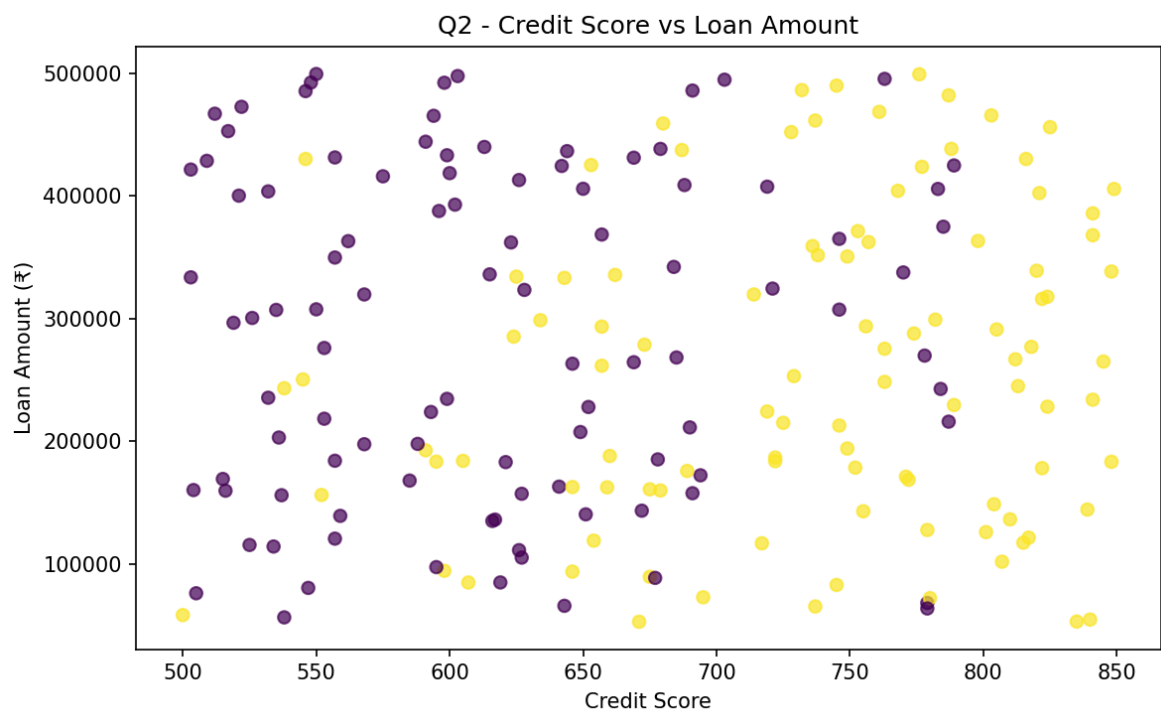


Figure 4. Credit Score vs Loan Amount

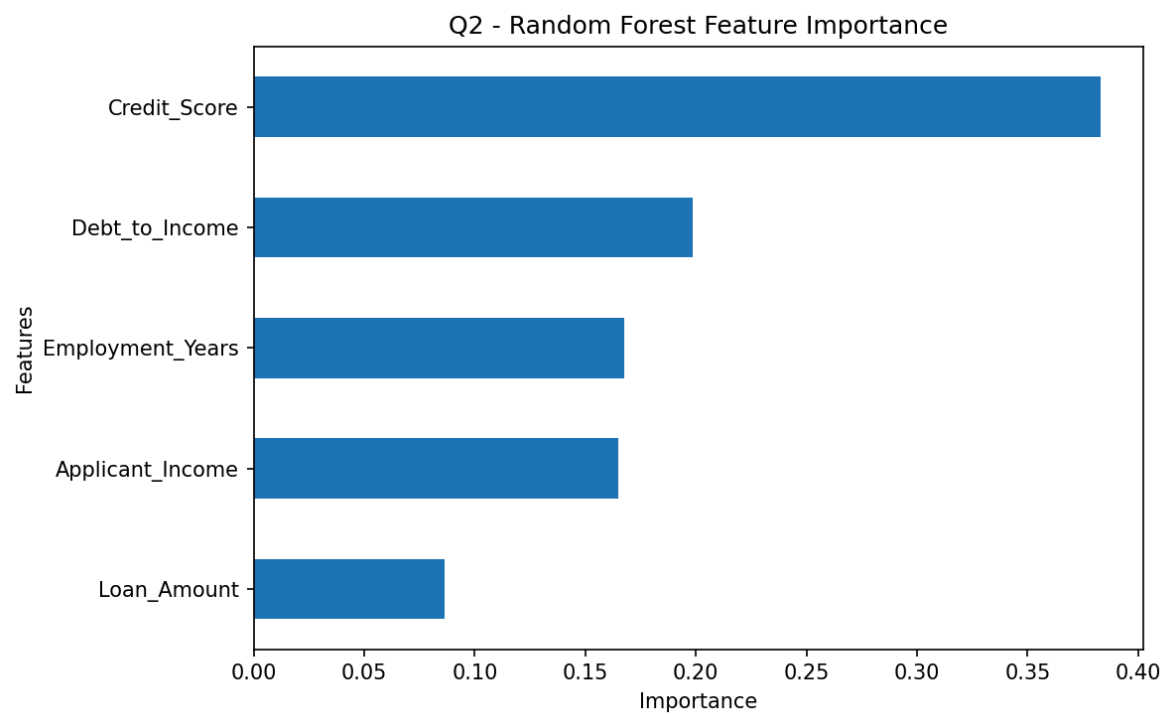


Figure 5. Random Forest Feature Importance

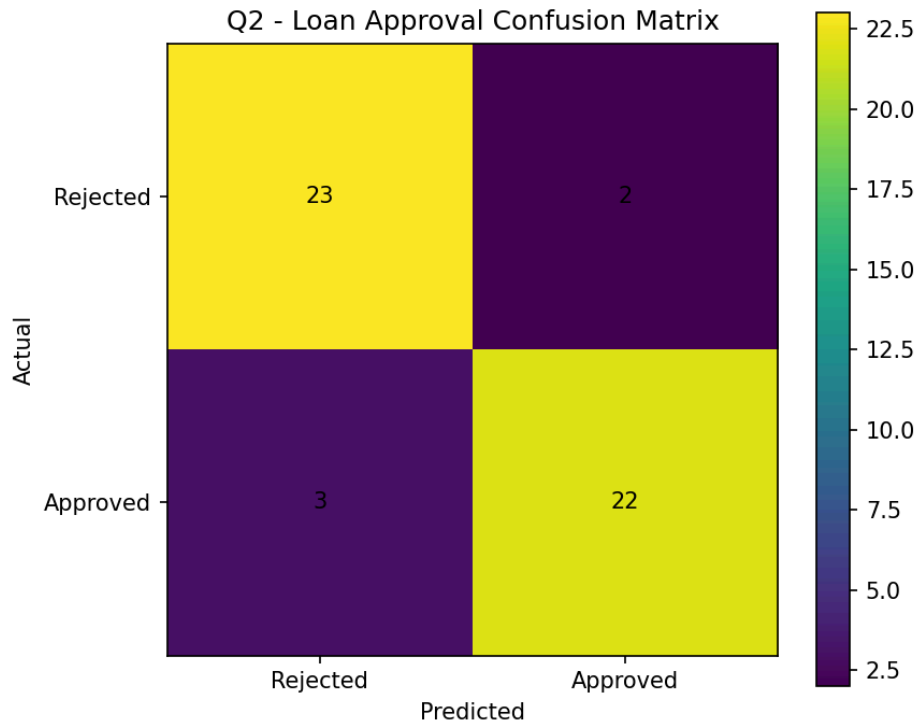


Figure 6. Loan Approval Confusion Matrix

Evaluation Values

Parameter	Value
TP	22
TN	23
FP	2
FN	3
Accuracy	90.00%
Precision	91.67%
Recall	88.00%
F1 Score	89.80%

Calculation of Parameters

Accuracy = $(22 + 23) / (22 + 23 + 2 + 3) = 45 / 50 = 0.90$

Accuracy = 90.00%

Precision = $22 / (22 + 2) = 22 / 24 = 0.9167$

Precision = 91.67%

Recall = $22 / (22 + 3) = 22 / 25 = 0.88$

Recall = 88.00%

F1 Score = $2 \times (0.9167 \times 0.88) / (0.9167 + 0.88) = 0.898$

F1 Score = 89.80%

Reliability, Limitations and Business Risks

Visualization	Reliability	Limitation	Business Risk
Credit Score vs Loan Amount	Shows applicant distribution and patterns	Uses only two variables	Wrong conclusions about credit risk
Feature Importance	Shows variables used strongly by the model	Importance is not causation	Management may act on misleading drivers
Confusion Matrix	Directly shows prediction errors	Does not show financial cost	Incorrect approvals can cause financial loss

Decision Framework

- Check dataset quality and representativeness.
- Evaluate Accuracy, Precision, Recall and F1 Score.
- Check whether the three visualizations give consistent evidence.
- Measure the financial cost of False Positives and False Negatives.
- Test the model on unseen/new data.
- Check model stability across different samples.
- Perform cost-benefit analysis.
- Use a small pilot before committing the full investment.

Conclusion

The model shows good predictive performance, but accuracy alone is not sufficient for a multi-million-rupee investment. A controlled pilot followed by further validation and financial risk analysis is recommended.

QUESTION 3 – Exercise Frequency and Recovery

A healthcare research team analyzes a large patient dataset and observes a strong positive relationship between the frequency of physical exercise and the recovery rate of patients suffering from chronic illnesses. Based on this observation alone, policymakers decide to invest substantial public funds in a nationwide fitness campaign, assuming that increased exercise will directly improve recovery outcomes across the population. Critically evaluate the validity of this decision.

Objective

To analyze the relationship between exercise frequency and recovery using a healthcare dataset and to determine whether the observed relationship is sufficient to establish causality.

Problem Statement

A positive relationship between exercise and recovery may be influenced by other variables. The analysis therefore examines hidden variables, causality, sampling bias, data quality, and the social and economic consequences of a nationwide policy.

Dataset

Feature	Description
Age	Patient age
Exercise_Frequency	Exercise sessions per week
Treatment_Adherence	Percentage of treatment adherence
Disease_Severity	Disease severity from 1 to 10
Diet_Score	Diet quality score
Recovery	1 = Recovered, 0 = Not Recovered

Machine Learning Model

Random Forest Classifier is used to predict recovery using all available patient factors.

Python Code

```
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (
    accuracy_score, precision_score,
    recall_score, f1_score, confusion_matrix
)

df = pd.read_csv("chronic_illness_recovery_dataset.csv")

X = df.drop("Recovery", axis=1)
y = df["Recovery"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25,
    random_state=42, stratify=y
)

model = RandomForestClassifier(
    n_estimators=100, random_state=42
)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)

print("Accuracy :", round(accuracy, 4))
print("Precision:", round(precision, 4))
print("Recall    :", round(recall, 4))
print("F1 Score  :", round(f1, 4))

cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:")
```

Visualizations

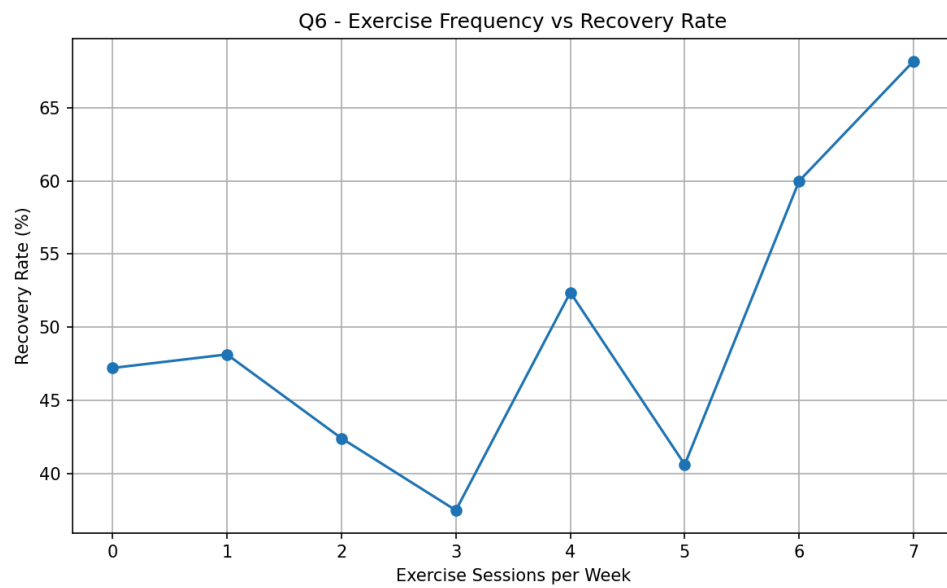


Figure 7. Exercise Frequency vs Recovery Rate

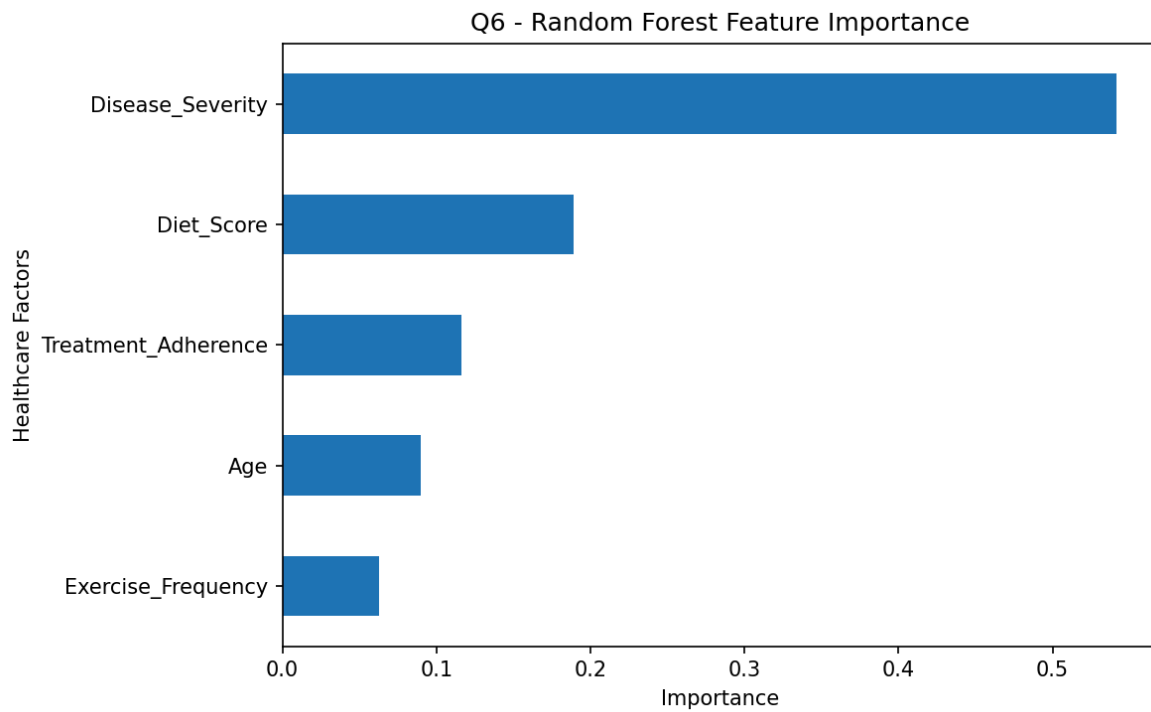


Figure 8. Random Forest Feature Importance

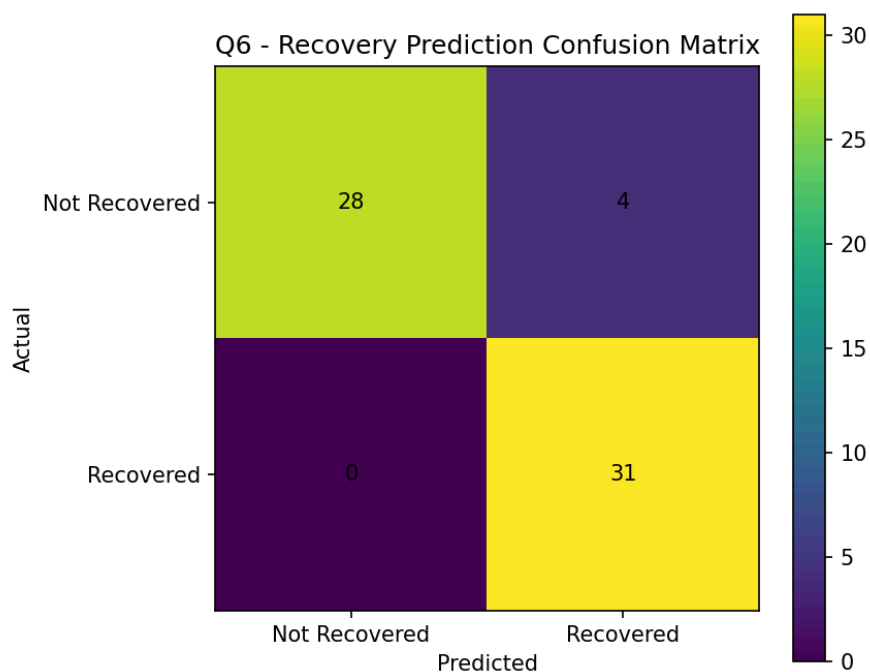


Figure 9. Recovery Prediction Confusion Matrix

Evaluation Values

Parameter	Value
TP	31
TN	28
FP	4
FN	0
Accuracy	93.65%
Precision	88.57%
Recall	100.00%
F1 Score	93.94%

Calculation of Parameters

Accuracy = $(31 + 28) / (31 + 28 + 4 + 0) = 59 / 63 = 0.9365$

Accuracy = 93.65%

Precision = $31 / (31 + 4) = 31 / 35 = 0.8857$

Precision = 88.57%

Recall = $31 / (31 + 0) = 1.00$

Recall = 100.00%

F1 Score = $2 \times (0.8857 \times 1.00) / (0.8857 + 1.00) = 0.9394$

F1 Score = 93.94%

Critical Evaluation

- Correlation does not prove causation. A positive exercise-recovery relationship may not mean exercise directly causes recovery.
- Hidden variables such as age, disease severity, diet, medication, treatment adherence and healthcare access may affect recovery.
- Reverse causality is possible: healthier or recovering patients may be more capable of exercising.
- Sampling bias can occur if the dataset is not representative of different ages, regions, income groups or disease types.
- Self-reported exercise, missing values and inconsistent recovery measurements can reduce data quality.
- Regression, longitudinal analysis and controlled clinical studies should be used before making a nationwide decision.
- A nationwide policy could have social and economic costs if the assumption is incorrect.

Conclusion

The observed relationship is useful evidence for further research, but it is not sufficient proof that exercise causes recovery. A controlled pilot and further statistical and medical investigation should precede a nationwide campaign.

QUESTION 4 – ₹500 Crore Public Health Investment

A nation plans to allocate ₹500 crore toward a public health initiative after observing a strong relationship between physical activity levels and patient recovery outcomes in a healthcare dataset. As a data scientist, evaluate whether the available evidence is sufficient to justify such an investment. Discuss the analytical, statistical, ethical, and policy-related factors that should be examined before translating data-driven observations into large-scale governmental action.

Objective

To evaluate whether the available healthcare evidence is strong enough to justify a ₹500 crore public health investment and to identify the analytical, statistical, ethical and policy checks required before nationwide implementation.

Problem Statement

A strong relationship between physical activity and recovery is observed, but the government must determine whether this relationship represents a causal and cost-effective effect before allocating ₹500 crore.

Dataset

The same Chronic Illness Recovery Dataset is used for demonstration. It contains 250 patient records and the variables Age, Exercise_Frequency, Treatment_Adherence, Disease_Severity, Diet_Score and Recovery.

Machine Learning Model

Random Forest Classifier is used to predict patient recovery. The model provides predictive evidence, but prediction performance does not by itself establish a causal effect of exercise.

Python Code

```
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (
    accuracy_score, precision_score,
    recall_score, f1_score, confusion_matrix
)

df = pd.read_csv("chronic_illness_recovery_dataset.csv")
```

```

X = df.drop("Recovery", axis=1)
y = df["Recovery"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25,
    random_state=42, stratify=y
)

model = RandomForestClassifier(
    n_estimators=100, random_state=42
)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)

print("Accuracy :", round(accuracy, 4))
print("Precision:", round(precision, 4))
print("Recall    :", round(recall, 4))
print("F1 Score  :", round(f1, 4))

cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:")
print(cm)

```

Visualizations

Q6 - Exercise Frequency vs Recovery Rate

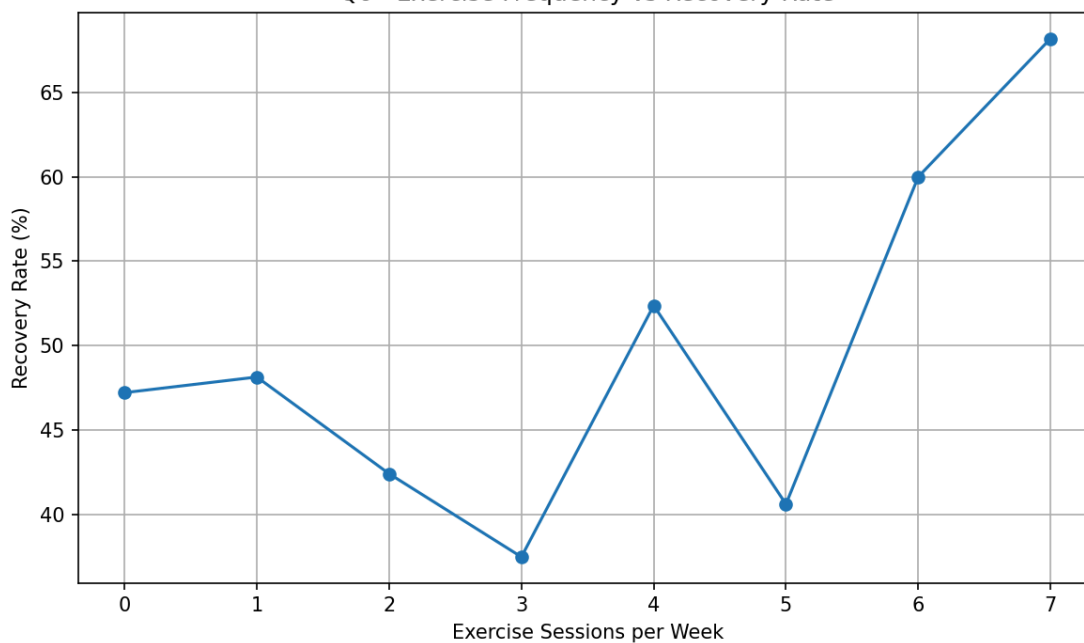


Figure 10. Exercise Frequency vs Recovery Rate

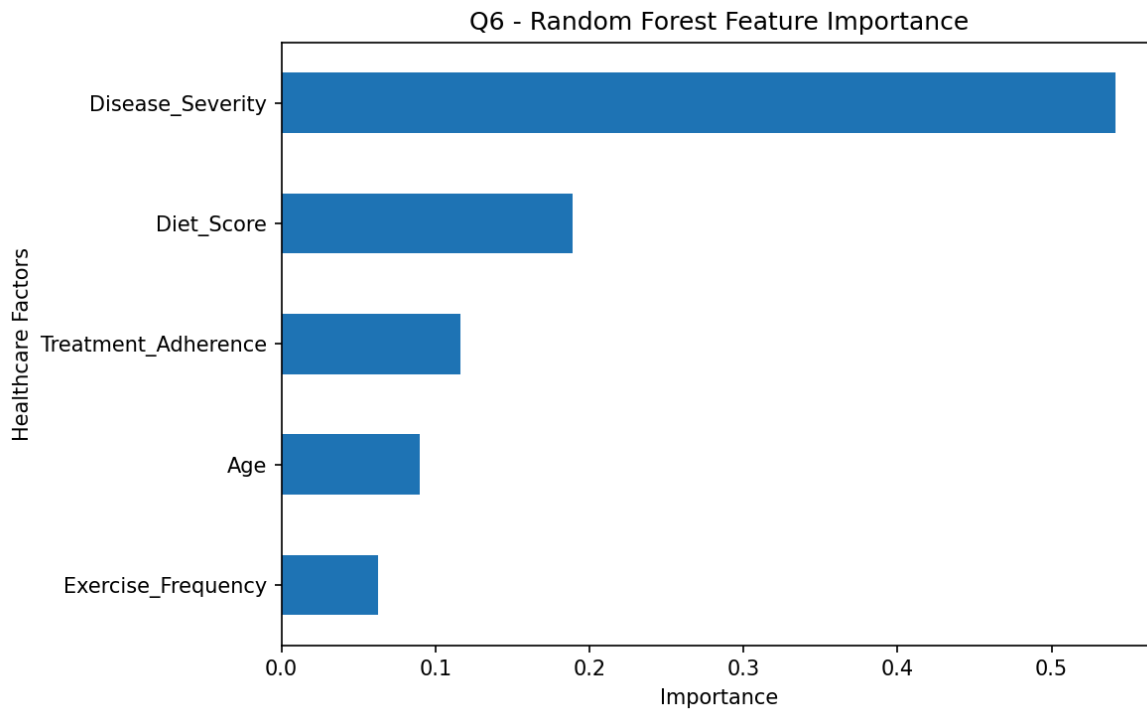


Figure 11. Random Forest Feature Importance

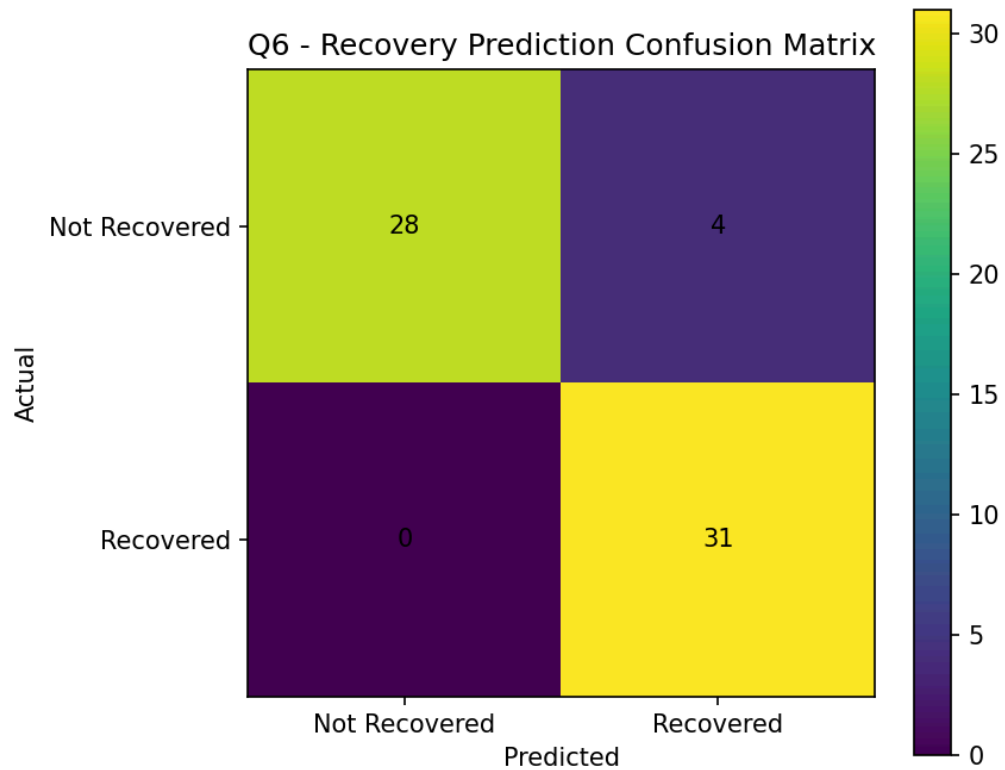


Figure 12. Recovery Prediction Confusion Matrix

Evaluation Values

Parameter	Value
TP	31
TN	28
FP	4
FN	0
Accuracy	93.65%
Precision	88.57%
Recall	100.00%
F1 Score	93.94%

Calculation of Parameters

Accuracy = $(31 + 28) / (31 + 28 + 4 + 0) = 59 / 63 = 0.9365 = 93.65\%$

Precision = $31 / (31 + 4) = 31 / 35 = 0.8857 = 88.57\%$

Recall = $31 / (31 + 0) = 1.00 = 100.00\%$

F1 Score = $2 \times (0.8857 \times 1.00) / (0.8857 + 1.00) = 0.9394 = 93.94\%$

Factors Before the ₹500 Crore Decision

Area	What Should Be Examined?
Analytical	Data quality, representativeness, missing data, model performance and visualization consistency
Statistical	Correlation, regression, confidence intervals, confounding variables and causal evidence
Clinical	Safety, suitable exercise intensity, disease-specific effects and controlled trials
Ethical	Privacy, fairness, accessibility and protection of vulnerable groups
Policy	Cost-benefit, implementation feasibility, regional differences and opportunity cost
Economic	Expected health benefit compared with ₹500 crore cost and possible unintended costs

Recommended Policy Framework

- Validate the dataset and check whether it represents the national population.
- Conduct statistical analysis and control for important confounding variables.
- Perform clinical and experimental studies to investigate causality.
- Run a smaller pilot program before nationwide implementation.
- Measure health outcomes, safety, equity and economic impact.
- Perform a cost-benefit analysis of the ₹500 crore investment.
- Scale nationally only if evidence supports effectiveness, safety and value for money.

Final Decision

The available relationship and model results are not sufficient by themselves to justify an immediate ₹500 crore allocation. The appropriate decision is to conduct further validation and a controlled pilot first. Large-scale implementation should occur only when causal evidence, safety, equity and cost-effectiveness are adequately demonstrated.